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A mixed finite element method for a nonlinear Dirichlet problem

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We study a mixed finite element approximation of a nonlinear Dirichlet problem in both two and three dimensions. This study is a first step towards the treatment of Ladyzhenskaya flows or quasi-Newtonian flows obeying the power law by mixed finite element methods. We give existence and uniqueness results for the continuous problem and its approximation and we prove an error bound.

1. Introduction

In this paper we study the finite element approximation of the following nonlinear Dirichlet problem:

$$\begin{cases} -\nabla \cdot (|\nabla u|^{s-2}\nabla u) = f & \text{in } \Omega, \\ u = 0 & \text{on } \Gamma, \end{cases} \tag{1.1}$$

where $1 < s < \infty$ and Ω is a bounded open subset of $\mathbb{R}^N$ ($N = 2$ or 3) with a Lipschitz continuous boundary Γ.

Above, ∇u is the vector gradient of u; for a vector function v, the divergence of v is the scalar

$$\nabla \cdot v = \sum_{i=1}^{N} \frac{\partial v_i}{\partial x_i} \quad \text{and} \quad |\nabla u| = \left(\sum_{i=1}^{N} \left(\frac{\partial u}{\partial x_i} \right)^2 \right)^{1/2}.$$

One of the difficulties encountered in the approximation of quasi-Newtonian flows obeying the power law (cf. Baranger & Najib (1990) and Sandri (1993)) and Ladyzhenskaya flows (cf. Ladyzhenskaya (1970)) is found in problem (1.1).

Analysis of finite element approximations of problem (1.1) and error bounds for this problem are known, see Glowinski & Marroco (1975), Ciarlet (1978), Tyukhtin (1983) and Barrett & Liu (1993). Further results for more general problems, including (1.1), are given in Liu and Barrett (1994, 1996), and the references therein. However in some applications $|\nabla u|^{s-2}\nabla u$ may be the more relevant variable. For example, when $s = 2$ (i.e., the classical Dirichlet problem), in thermodiffusion problems, ∇u will be the heat flux, which is very often more important to know than the temperature u. In such cases, the use of a mixed formulation of problem (1.1), by introducing the auxiliary variable $\sigma = |\nabla u|^{s-2}\nabla u$, is more appropriate as long as it provides better accuracy for σ.

The main purpose of this paper is to study a mixed formulation of problem (1.1) where $\sigma = |\nabla u|^{s-2}\nabla u$ is approximated by the lowest-degree Raviart–Thomas finite element (cf.

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Raviart & Thomas (1977)) and u by piecewise constants. This kind of approximation is a generalization of the mixed finite element method of Raviart & Thomas (1977) for the case $s = 2$ to the case $s \neq 2$.

The mixed formulation, considered in this paper, is based on the inversion of the relation $\sigma = |\nabla u|^{s-2}\nabla u$, i.e.,

$$\nabla u = |\sigma|^{s'-2}\sigma$$

where $\frac{1}{s} + \frac{1}{s'} = 1$, and (1.1) is expressed as follows:

$$\begin{cases} |\sigma|^{s'-2}\sigma = \nabla u & \text{in } \Omega, \\ \nabla \cdot \sigma + f = 0 & \text{in } \Omega, \\ u = 0 & \text{on } \Gamma. \end{cases} \tag{1.2}$$

We give existence and uniqueness results for the continuous problem (1.2) and its approximation and we prove an error bound in the case $1 < s < 2$. The case $s > 2$ presents some technical difficulties. For $s = 2$, we refer to Raviart & Thomas (1977) for the case $N = 2$, and to Brezzi & Fortin (1991) and Roberts & Thomas (1989) for the case $N = 3$.

Let us mention that the solution u of problem (1.1) is not, in general, very regular even with very smooth data (cf. Glowinski & Marroco (1975)). This is why we consider the approximation of σ only by the lowest-order Raviart–Thomas finite element and not by higher-order ones. We also mention that this work is a first step towards the treatment of Ladyzhenskaya flows or quasi-Newtonian flows obeying the power law by a mixed finite element method similar to the one developed here.

2. The mixed formulation

In order to derive the mixed formulation of problem (1.2), we first define the following spaces

$$X = \left\{\tau \in (L^{s'}(\Omega))^N;\ \nabla \cdot \tau \in L^2(\Omega)\right\}, \qquad M = L^2(\Omega). \tag{2.1}$$

We denote the norm of X by $\|\tau\|_X = \|\tau\|_{0,s',\Omega} + \|\nabla \cdot \tau\|_{0,\Omega}$, where $\|\tau\|_{0,s',\Omega} = \left(\int_\Omega |\tau|^{s'} \,\mathrm{d}x\right)^{1/s'}$.

Setting

$$A(\sigma) = |\sigma|^{s'-2}\sigma, \tag{2.2}$$

the mixed formulation of (1.2) reads as follows: Find $\sigma \in X$ and $u \in M$ such that

$$\begin{cases} \displaystyle\int_\Omega A(\sigma) \cdot \tau \,\mathrm{d}x + \int_\Omega \nabla \cdot \tau\, u \,\mathrm{d}x = 0 \ \ \forall \tau \in X, \\ \displaystyle\int_\Omega \nabla \cdot \sigma\, v \,\mathrm{d}x + \int_\Omega f\, v \,\mathrm{d}x = 0 \ \ \forall v \in M, \end{cases} \tag{2.3}$$

where $f \in L^2(\Omega)$.

Subsequently, we suppose that $1 < s < 2$.

To prove the existence and the uniqueness of the solution of (2.3) we use minimization techniques as has been done by Baranger *et al* (1993) for a quasi-Newtonian flow. Before doing that, let us prove the following result concerning the inf-sup condition.

PROPOSITION 2.1 There exists a positive constant β such that

$$\inf_{v\in M}\sup_{\tau\in X}\frac{\int_\Omega \nabla\cdot\tau\, v\,dx}{\|\tau\|_X\|v\|_{0,\Omega}}\geqslant\beta. \tag{2.4}$$

Proof. Let $v\in M$ and $\tau^*=-|\nabla w|^{s-2}\nabla w$, where w is the solution of the nonlinear Dirichlet problem

$$\begin{cases} -\nabla\cdot(|\nabla w|^{s-2}\nabla w)=v & \text{in } \Omega,\\ w=0 & \text{on } \Gamma.\end{cases}$$

Recall that (cf. Glowinski & Marroco (1975)) the problem above has a unique solution $w\in W_0^{1,s}(\Omega)$ and there exists a constant C such that

$$\|w\|_{1,s,\Omega}\leqslant C\,\|v\|_{0,\Omega}^{1/(s-1)}.$$

Therefore, $\tau^*\in X$, there exists a constant C such that $\|\tau^*\|_X\leqslant C\,\|v\|_{0,\Omega}$, and $\int_\Omega \nabla\cdot\tau^*\,v\,dx=\|v\|_{0,\Omega}^2$. Thus, (2.4) follows immediately. □

We also need the following abstract result (see Remark 4.2, p 61 of Girault & Raviart (1986), and Sandri (1993)).

LEMMA 2.1 Let $(X,\|\cdot\|_X)$ and $(M,\|\cdot\|_M)$ be two reflexive Banach spaces. Let $(X',\|\cdot\|_{X'})$ and $(M',\|\cdot\|_{M'})$ be their corresponding dual space. Let $B: X\to M'$ be a linear continuous operator and $B': M\to X'$ the dual operator of B. Let $V=\ker(B)$ be the kernel of B; we denote by $V^o\subset X'$ the polar set of V, i.e., $V^o=\{x'\in X',\ \langle x',v\rangle=0,\ \forall v\in V\}$, and $\dot B: (X/V)\to M'$ the quotient operator associated with B. The three following properties are equivalent:

(i) there exists $\beta>0$, such that

$$\inf_{q\in M}\sup_{v\in X}\frac{\langle Bv,q\rangle}{\|q\|_M\|v\|_X}\geqslant\beta,$$

(ii) B' is an isomorphism from M onto V^o and

$$\|B'q\|_{X'}\geqslant\beta\|q\|_M\quad\forall q\in M,$$

(iii) $\dot B$ is an isomorphism from (X/V) onto M' and

$$\|\dot B\dot v\|_{M'}\geqslant\beta\|\dot v\|_{(X/V)}\quad\forall\dot v\in(X/V).$$

Now, we have the following result.

THEOREM 2.1 Problem (2.3) admits a unique solution satisfying

$$\|\sigma\|_X+\|u\|_{0,\Omega}\leqslant C\left(\|f\|_{0,\Omega}+\|f\|_{0,\Omega}^{s'-1}\right),$$

where C is a positive constant.

Proof. Let $K(f) = \{\tau \in X;\ (\nabla\cdot\tau, v)+(f, v) = 0,\ \forall v \in M\}$ and let $B = \nabla\cdot$ defined from X onto M. Using the inf-sup condition (2.4) and applying Lemma 2.1(iii) to the operator B, we get the existence of $\sigma_0 \in K(f)$ such that $\|\sigma_0\|_X \leqslant (1/\beta)\|f\|_{0,\Omega}$. Therefore, problem (2.3) may be written in the following way:

$$\begin{cases} \text{Find } \hat{\sigma} = (\sigma - \sigma_0) \in X \text{ and } u \in M \text{ such that} \\ \displaystyle\int_\Omega A(\hat{\sigma} + \sigma_0)\cdot\tau\,dx + \int_\Omega \nabla\cdot\tau\, u\,dx = 0 \ \ \forall\tau \in X \\ \displaystyle\int_\Omega \nabla\cdot\hat{\sigma}\, v\,dx = 0 \ \ \forall v \in M \end{cases} \tag{2.5}$$

or

$$\begin{cases} \text{Find } \hat{\sigma} \in K(0) \text{ such that} \\ \displaystyle\int_\Omega A(\hat{\sigma} + \sigma_0)\cdot\tau\,dx = 0 \ \ \forall\tau \in K(0). \end{cases} \tag{2.6}$$

Now, let J be the functional defined by

$$J(\tau) = \frac{1}{s'}\int_\Omega |\tau|^{s'}\,dx \quad \forall\tau \in (L^{s'}(\Omega))^N.$$

Then problem (2.6) is equivalent to

$$\begin{cases} \text{Find } \hat{\sigma} \in K(0) \text{ such that} \\ J(\hat{\sigma} + \sigma_0) = \inf\limits_{\tau\in K(0)} J(\tau + \sigma_0). \end{cases} \tag{2.7}$$

From the properties of J (cf. Glowinski & Marroco (1975), Ciarlet (1978)), one can deduce that problem (2.7) has a unique solution $\hat{\sigma}$. Furthermore,

$$\|\sigma\|_{0,s',\Omega} = (s'J(\hat{\sigma} + \sigma_0))^{1/s'} \leqslant (s'J(\sigma_0))^{1/s'} = \|\sigma_0\|_{0,s',\Omega} \leqslant (1/\beta)\|f\|_{0,\Omega},$$

and since $\nabla\cdot\sigma = -f$, we obtain $\|\sigma\|_X \leqslant (1 + 1/\beta)\|f\|_{0,\Omega}$.

On the other hand, using once again the inf-sup condition (2.4) and Lemma 2.1(ii), we obtain the existence of u such that (σ, u) satisfies (2.5), $\|u\|_{0,\Omega} \leqslant (1/\beta)\|\sigma\|_X^{s'-1}$, and the result follows immediately. □

REMARK 2.1 One can use the fact that $(\sigma = |\nabla u|^{s-2}\nabla u, u)$, where u is the solution of problem (1.1), is a solution of (2.3) and use the inf-sup condition (2.4) to prove the uniqueness of the solution of problem (2.3). However, the proof given above will be useful to prove the existence and the uniqueness of the solution of the discrete problem (see Section 3).

3. The discrete problem and error analysis

We assume that the boundary Γ of Ω is polygonal or polyhedral and we denote by T_h a triangulation of $\overline{\Omega}$ into closed triangles (for the case $N = 2$) or tetrahedrons (for the case $N = 3$). The elements of T_h are assumed to be regular in the usual sense (cf. Ciarlet (1978)).

Our finite element approximation of problem (2.3) is defined as follows: Find $\sigma_h \in X_h$ and $u_h \in M_h$ such that

$$\begin{cases} \displaystyle\int_\Omega A(\sigma_h)\cdot\tau_h\,dx + \int_\Omega \nabla\cdot\tau_h\,u_h\,dx = 0 \ \ \forall\tau_h \in X_h, \\ \displaystyle\int_\Omega \nabla\cdot\sigma_h\,v_h\,dx + \int_\Omega f\,v_h\,dx = 0 \ \ \forall v_h \in M_h, \end{cases} \tag{3.1}$$

where the finite element spaces are defined through

$$X_h = \{\tau_h \in X;\ \tau_h|_K \in RT_0(K),\ \forall K \in T_h\}, \tag{3.2}$$

$$M_h = \{v_h \in L^2(\Omega);\ v_h|_K \in P_0(K),\ \forall K \in T_h\}, \tag{3.3}$$

with $P_0(K)$ denoting the space of constants and

$$RT_0(K) = (P_0(K))^N + x\,P_0(K),\quad x = (x_1, \ldots, x_N).$$

The finite element space X_h is simply the approximation of the space X by the lowest-degree Raviart–Thomas element (cf. Raviart & Thomas (1977), Nedelec (1980), Brezzi & Fortin (1991), Roberts & Thomas (1989)). Then there exists an operator Π_h defined from X onto X_h such that, for all $K \in T_h$,

$$\int_{\partial K} (\tau - \Pi_K\tau)\cdot n\,p_0\,ds = 0 \ \ \forall p_0 \in R_0(\partial K), \tag{3.4}$$

where $\Pi_K\tau = \Pi_h\tau|_K$, n denotes the unit outward normal to the boundary ∂K of K, and

$$R_0(\partial K) = \{\psi;\ \psi \in P_0(e)\ \ \forall e \in \partial K\}.$$

REMARK 3.1 The operator Π_h is well defined for all $\tau \in X$ since $s' > 2$.

Before giving the error estimates for problem (3.1), let us first prove some intermediate results.

LEMMA 3.1 For all $\tau \in (W^{1,q}(\Omega))^N$, $q \geqslant 2$,

$$\|\tau - \Pi_h\tau\|_{0,q,\Omega} \leqslant Ch|\tau|_{1,q,\Omega}, \tag{3.5}$$

where C is a constant independent of h.

Proof. Let $\widehat{K}$ denote the reference unit simplex in the $\widehat{x} = (\widehat{x}_1, \ldots, \widehat{x}_N)$ space, where $N = 2$ or 3. For each $K \in T_h$, there exists an affine transformation $F_K : \widehat{K} \longrightarrow K = F_K(\widehat{K})$, bijective from $\widehat{K}$ onto K. We denote by DF_K the linear tangent mapping and by J the Jacobian of the transformation. To each vector function $\widehat{\tau}$, defined on $\widehat{K}$, we associate the function τ by using the so-called Piola transformation, i.e.,

$$\tau(x) = \frac{1}{|J|} DF_K\widehat{\tau}(\widehat{x}).$$

Now let $RT_0(\hat{K}) = (P_0(\hat{K}))^N + \hat{x} P_0(\hat{K})$ and let $\hat{\Pi}$ be the operator defined from $\{\hat{\tau} \in (L^{s'}(\hat{K}))^N;\ \nabla \cdot \hat{\tau} \in L^2(\hat{K})\}$ onto $RT_0(\hat{K})$ by

$$\int_{\partial \hat{K}} (\hat{\tau} - \hat{\Pi}\hat{\tau}) \cdot \hat{n}\, \hat{p}_0 \, \mathrm{d}\hat{s} = 0 \quad \forall \hat{p}_0 \in R_0(\partial \hat{K}).$$

Then, it is known that (cf. Girault & Raviart (1986), Roberts & Thomas (1989))

$$\Pi_h \tau|_K \circ F_K = \frac{1}{|J|} DF_K(\hat{\Pi}\hat{\tau}) \quad \forall K \in T_h.$$

On the other hand, since $(P_0(\hat{K}))^N$ is invariant under the interpolation operator $\hat{\Pi}$, there exists a constant $\hat{C}$ depending only on $\hat{K}$ and q such that, for all $q \geqslant 2$ and all $\hat{\tau} \in (W^{1,q}(\hat{K}))^N$,

$$\|\hat{\tau} - \hat{\Pi}\hat{\tau}\|_{0,q,\hat{K}} \leqslant \hat{C}\, |\hat{\tau}|_{1,q,\hat{K}}.$$

With this bound and the following formulas (cf. Thomas (1977))

$$|\tau|_{m,q,K} \leqslant C|J|^{1/q-1} \|DF_K\|\, \|DF_K^{-1}\|^m |\hat{\tau}|_{m,q,\hat{K}},$$

$$|\hat{\tau}|_{m,q,\hat{K}} \leqslant C|J|^{1-1/q} \|DF_K^{-1}\|\, \|DF_K\|^m |\tau|_{m,q,K},$$

we obtain, for all $K \in T_h$,

$$\|\tau - \Pi_h \tau\|_{0,q,K} \leqslant C\, h_K\, |\tau|_{1,q,K},$$

where h_K is the diameter of K.

Therefore, (3.5) follows immediately. □

PROPOSITION 3.1 There exists a constant C independent of h such that

$$\inf_{v_h \in M_h} \sup_{\tau_h \in X_h} \frac{\int_\Omega \nabla \cdot \tau_h\, v_h \, \mathrm{d}x}{\|\tau_h\|_X\, \|v_h\|_{0,\Omega}} \geqslant C. \tag{3.6}$$

Proof. Let $v_h \in M_h$. There exists (cf. Girault & Raviart (1986), Roberts & Thomas (1989)) $\tau_h^* \in (L^{s'}(\Omega))^N$ such that $\tau_h^*|_K \in RT_0(K), \forall K \in T_h$, $\nabla \cdot \tau_h^* = v_h$ in Ω and

$$\|\tau_h^*\|_{0,s',\Omega} + \|\nabla \cdot \tau_h^*\|_{0,\Omega} \leqslant C\|v_h\|_{0,\Omega}.$$

Then, $\tau_h^* \in X_h$, $\|\tau_h^*\|_X \leqslant C\|v_h\|_{0,\Omega}$, and

$$\int_\Omega \nabla \cdot \tau_h^*\, v_h \, \mathrm{d}x = \|v_h\|_{0,\Omega}^2.$$

Therefore, (3.6) follows immediately. □

We are now in a position to state the error estimates for problem (3.1).

THEOREM 3.1 Problem (3.1) has a unique solution $(\sigma_h, u_h) \in X_h \times M_h$. Further, if the solution (σ, u) of problem (2.3) satisfies $(\sigma, u) \in (W^{1,s'}(\Omega))^N \times H^1(\Omega)$, then there exists a constant C, independent of h, such that

$$\|\sigma - \sigma_h\|_{0,s',\Omega} \leqslant C\, h^{s-1}, \quad \|u - u_h\|_{0,\Omega} \leqslant C\, h^{s-1}. \tag{3.7}$$

Proof. Existence and uniqueness are obtained as in Theorem 2.1.

To prove the error estimate for σ, first observe that, from the second equation of (2.3) and of (3.1) respectively and the properties of the operator Π_h,

$$\int_\Omega \nabla \cdot (\Pi_h \sigma - \sigma_h)\, v_h \, dx = 0 \quad \forall v_h \in M_h.$$

On the other hand, by using the first equation of (2.3) and of (3.1) respectively and this last relation, we get

$$\int_\Omega (A(\sigma) - A(\sigma_h)) \cdot (\sigma_h - \Pi_h \sigma)\, dx = 0,$$

and then

$$\int_\Omega (A(\sigma_h) - A(\Pi_h \sigma)) \cdot (\sigma_h - \Pi_h \sigma)\, dx = \int_\Omega (A(\sigma) - A(\Pi_h \sigma)) \cdot (\sigma_h - \Pi_h \sigma)\, dx. \quad (3.8)$$

Now, using the fact that (cf. Glowinski & Marroco (1975))

$$\int_\Omega (A(\sigma_h) - A(\Pi_h \sigma)) \cdot (\sigma_h - \Pi_h \sigma)\, dx \geqslant \|\sigma_h - \Pi_h \sigma\|^{s'}_{0,s',\Omega},$$

$$\|A(\sigma) - A(\Pi_h \sigma)\|_{0,s,\Omega} \leqslant C\, (\|\sigma\|_{0,s',\Omega} + \|\Pi_h \sigma\|_{0,s',\Omega})^{s'-2}\, \|\sigma - \Pi_h \sigma\|_{0,s',\Omega},$$

and (3.8), we get

$$\|\sigma_h - \Pi_h \sigma\|^{s'-1}_{0,s',\Omega} \leqslant C\, (\|\sigma\|_{0,s',\Omega} + \|\Pi_h \sigma\|_{0,s',\Omega})^{s'-2}\, \|\sigma - \Pi_h \sigma\|_{0,s',\Omega}.$$

This last estimate, with (3.5), gives us

$$\|\sigma_h - \Pi_h \sigma\|_{0,s',\Omega} \leqslant C\, h^{1/(s'-1)} |\sigma|^{1/(s'-1)}_{1,s',\Omega} = C\, h^{s-1} |\sigma|^{s-1}_{1,s',\Omega},$$

where C is a constant independent of h. Therefore, using the triangle inequality and (3.5), we get the first error estimate of (3.7).

It remains to prove the error estimate for u. First observe that, from the first equation of (2.3) and of (3.1) respectively

$$\int_\Omega \nabla \cdot \tau_h\, (P_h u - u_h)\, dx = \int_\Omega (A(\sigma_h) - A(\sigma)) \cdot \tau_h \, dx \quad \forall \tau_h \in X_h,$$

where $P_h u$ is the L^2-projection of u onto $\prod_{K \in T_h} P_0(K)$.

Thus, using this last relation and the inf-sup condition (3.6), we get

$$\begin{aligned} C\, \|P_h u - u_h\|_{0,\Omega} &\leqslant \sup_{\tau_h \in X_h} \frac{\int_\Omega \nabla \cdot \tau_h\, (P_h u - u_h)\, dx}{\|\tau_h\|_X} \\ &\leqslant \sup_{\tau_h \in X_h} \frac{\int_\Omega (A(\sigma_h) - A(\sigma)) \cdot \tau_h \, dx}{\|\tau_h\|_X}. \end{aligned}$$

Therefore,

$$\|P_h u - u_h\|_{0,\Omega} \leqslant C\, \|A(\sigma) - A(\sigma_h)\|_{0,s,\Omega} \leqslant C\, \|\sigma - \sigma_h\|_{0,s',\Omega} \quad (3.9)$$

owing to the fact that (cf. Glowinski & Marroco (1975))

$$\|A(\sigma) - A(\sigma_h)\|_{0,s,\Omega} \leqslant C\,(\|\sigma\|_{0,s',\Omega} + \|\sigma_h\|_{0,s',\Omega})^{s'-2}\,\|\sigma - \sigma_h\|_{0,s',\Omega}.$$

Finally, the second error estimate of (3.7) is a consequence of the one for σ, (3.9), the triangle inequality and the following standard error estimation

$$\|u - P_h u\|_{0,\Omega} \leqslant C\,h\,|u|_{1,\Omega}.$$

□

REMARK 3.2 In the proof of Theorem 3.1, we have used the fact that $\|\sigma_h\|_{0,s',\Omega} \leqslant C$ with C a constant independent of h. The proof of this bound is analogous to the one for σ in Theorem 2.1, owing to the inf-sup condition (3.6).

The error estimation (3.7) for σ is the same as the one obtained in Baranger *et al* (1993) for a quasi-Newtonian flow, but the error estimation for u is not optimal. One can improve these error estimates under some regularity on the solution of problem (2.3). To this end, we need the following result.

LEMMA 3.2 Suppose that $1 < s < 2$ in the two-dimensional case ($N = 2$), and $6/5 \leqslant s < 2$ in the three-dimensional case ($N = 3$). Then, the operator Π_h, defined by (3.4), satisfies

$$\int_\Omega \nabla \cdot (\Pi_h \tau - \tau)\, v_h \,dx = 0,\ \forall \tau \in X,\ \forall v_h \in M_h, \tag{3.10}$$

$$\|\Pi_h \tau\|_X \leqslant C\,\|\tau\|_X,\ \forall \tau \in X, \tag{3.11}$$

where C is a constant independent of h.

Proof. First, recall that, for all $\tau \in (H^1(\Omega))^N$,

$$\int_\Omega \nabla \cdot (\Pi_h \tau - \tau)\, v_h \,dx = 0,\ \forall v_h \in M_h.$$

On the other hand, using the same arguments as in the proof of Lemma 3.1, we have

$$\|\tau - \Pi_h \tau\|_{0,s',\Omega} \leqslant \begin{cases} Ch^{2/s'}|\tau|_{1,\Omega},\ \forall \tau \in (H^1(\Omega))^2,\ \forall s' \geqslant 2, \\ Ch^{3/s'-1/2}|\tau|_{1,\Omega},\ \forall \tau \in (H^1(\Omega))^3,\ \forall s',\ 2 \leqslant s' \leqslant 6, \end{cases} \tag{3.12}$$

where C is a constant independent of h.

The condition on s' ($2 \leqslant s' \leqslant 6$) in the three-dimensional case comes from the fact that, for a bounded open subset Ω of $\mathbb{R}^3$, the imbedding of $H^1(\Omega)$ in $L^{q'}(\Omega)$ holds for $1 \leqslant q' \leqslant 6$.

Now, let $\tau \in X$ and $\tilde{\tau} = \nabla w$, where w is the solution of the Dirichlet problem

$$\begin{cases} \Delta w = \nabla \cdot \tau \ \text{ in } \Omega, \\ w = 0 \ \text{ on } \Gamma. \end{cases}$$

Since $\nabla \cdot \tau \in L^2(\Omega)$, we have $\tilde{\tau} \in (H^1(\Omega))^N$, there exists a constant C such that $\|\tilde{\tau}\|_{1,\Omega} \leqslant C\|\nabla \cdot \tau\|_{0,\Omega}$, and $\nabla \cdot \tilde{\tau} = \nabla \cdot \tau$ in Ω.

Therefore, setting $\Pi_h\tau = \Pi_h\tilde{\tau}$, we get

$$\int_\Omega \nabla\cdot\Pi_h\tau\, v_h\,dx = \int_\Omega \nabla\cdot\Pi_h\tilde{\tau}\, v_h\,dx = \int_\Omega \nabla\cdot\tilde{\tau}\, v_h\,dx = \int_\Omega \nabla\cdot\tau\, v_h\,dx,$$

for all $v_h \in M_h$.

It remains to prove (3.11). Using (3.12), we get

$$\|\Pi_h\tau\|_{0,s',\Omega} = \|\Pi_h\tilde{\tau}\|_{0,s',\Omega} \leqslant \|\Pi_h\tilde{\tau} - \tilde{\tau}\|_{0,s',\Omega} + \|\tilde{\tau}\|_{0,s',\Omega}$$

$$\leqslant \begin{cases} C(h^{2/s'}|\tilde{\tau}|_{1,\Omega} + \|\tilde{\tau}\|_{1,\Omega}),\ \forall s' > 2,\ N = 2, \\ C(h^{3/s'-1/2}|\tilde{\tau}|_{1,\Omega} + \|\tilde{\tau}\|_{1,\Omega}),\ \forall s', 2 < s' \leqslant 6,\ N = 3. \end{cases}$$

Thus, $\|\Pi_h\tau\|_{0,s',\Omega} \leqslant C\|\nabla\cdot\tau\|_{0,\Omega}$, and since $\|\nabla\cdot\Pi_h\tau\|_{0,\Omega} \leqslant C\|\nabla\cdot\tau\|_{0,\Omega}$, we obtain (3.11). □

REMARK 3.3 To have (3.10) and (3.11) in the three-dimensional case for all s, $1 < s < 2$, we believe that the definition of the space X has to be replaced by the following one

$$X = \{\tau \in (L^{s'}(\Omega))^N;\ \nabla\cdot\tau \in L^{s'}(\Omega)\}.$$

We now state the result about the improvement of the error estimation (3.7).

THEOREM 3.2 Suppose that $1 < s < 2$ in the two-dimensional case, and $6/5 \leqslant s < 2$ in the three-dimensional case. Let (σ, u) be the solution of (2.3) and (σ_h, u_h) the solution of (3.1). If $(\sigma, u) \in (W^{1,s'}(\Omega))^N \times H^1(\Omega)$ and $\nabla\cdot\sigma \in H^1(\Omega)$, then there exists a constant C independent of h such that

$$\|\sigma - \sigma_h\|_{0,s',\Omega} \leqslant C\,h^{2/s'}, \tag{3.13}$$

$$\|u - u_h\|_{0,\Omega} \leqslant C\,h. \tag{3.14}$$

Proof. Let us start with the proof of (3.13). We set

$$K_h(f) = \left\{\tau_h \in X_h;\ \int_\Omega \nabla\cdot\tau_h\, v_h\,dx + \int_\Omega f\, v_h\,dx = 0,\ \forall v_h \in M_h\right\}.$$

Let τ_h be in $K_h(f)$; from (2.3) and (3.1) we obtain

$$\int_\Omega (A(\sigma) - A(\sigma_h))\cdot(\tau_h - \sigma_h)\,dx = \int_\Omega \nabla\cdot(\tau_h - \sigma_h)\,(u_h - u)\,dx = 0,$$

and then

$$\int_\Omega (A(\sigma) - A(\sigma_h))\cdot(\sigma - \sigma_h)\,dx = \int_\Omega (A(\sigma) - A(\sigma_h))\cdot(\sigma - \tau_h)\,dx\ \ \forall\tau_h \in K_h(f).$$

From (cf. Sandri (1993))

$$\int_\Omega (A(\sigma)-A(\sigma_h))\cdot(\sigma-\sigma_h)\,dx \geqslant C\left\{\|\sigma - \sigma_h\|^{s'}_{0,s',\Omega} + \int_\Omega (|\sigma| + |\sigma_h|)^{s'-2}|\sigma - \sigma_h|^2\,dx\right\}, \tag{3.15}$$

$$\|A(\sigma) - A(\sigma_h)\|_{0,s,\Omega}$$

$$\leqslant C\left[\int_\Omega (|\sigma| + |\sigma_h|)^{s'-2}|\sigma - \sigma_h|^2\,dx\right]^{1/2} [\|\sigma\|_{0,s',\Omega} + \|\sigma_h\|_{0,s',\Omega}]^{(s'-2)/2}, \quad (3.16)$$

and using Young's inequality, we obtain, for all $\epsilon > 0$,

$$\|\sigma - \sigma_h\|_{0,s',\Omega}^{s'} + \int_\Omega (|\sigma| + |\sigma_h|)^{s'-2}|\sigma - \sigma_h|^2\,dx \leqslant \epsilon \int_\Omega (|\sigma| + |\sigma_h|)^{s'-2}|\sigma - \sigma_h|^2\,dx$$
$$+\frac{C}{\epsilon}[\|\sigma\|_{0,s',\Omega} + \|\sigma_h\|_{0,s',\Omega}]^{s'-2}\,\|\sigma - \tau_h\|_{0,s',\Omega}^2,$$

where C is a constant independent of h and ϵ. Using this last estimate and Remark 3.2, we get

$$\|\sigma - \sigma_h\|_{0,s',\Omega}^{s'} + \int_\Omega (|\sigma| + |\sigma_h|)^{s'-2}|\sigma - \sigma_h|^2\,dx \leqslant C \inf_{\tau_h \in K_h(f)} \|\sigma - \tau_h\|_{0,s',\Omega}^2. \quad (3.17)$$

Now let us show that

$$\inf_{\tau_h \in K_h(f)} \|\sigma - \tau_h\|_{0,s',\Omega} \leqslant C \inf_{\tau_h \in X_h} \|\sigma - \tau_h\|_X. \quad (3.18)$$

Let $\tau_h \in X_h$ and $\tilde{\sigma} = \tau_h + \Pi_h(\sigma - \tau_h)$. Owing to (3.10), we have, for all $v_h \in M_h$,

$$\int_\Omega \nabla \cdot \tilde{\sigma}\, v_h\,dx = \int_\Omega \nabla \cdot \tau_h\, v_h\,dx + \int_\Omega \nabla \cdot (\Pi_h(\sigma - \tau_h))\, v_h\,dx$$
$$= \int_\Omega \nabla \cdot \tau_h\, v_h\,dx + \int_\Omega \nabla \cdot (\sigma - \tau_h)\, v_h\,dx$$
$$= \int_\Omega \nabla \cdot \sigma\, v_h\,dx = -\int_\Omega f\, v_h\,dx$$

and then $\tilde{\sigma} \in K_h(f)$. Furthermore,

$$\|\tilde{\sigma} - \tau_h\|_{0,s',\Omega} = \|\Pi_h(\sigma - \tau_h)\|_{0,s',\Omega} \leqslant C\,\|\sigma - \tau_h\|_X, \quad \forall \tau_h \in X_h,$$

owing to (3.11). Therefore,

$$\inf_{\tau_h \in K_h(f)} \|\sigma - \tau_h\|_{0,s',\Omega} \leqslant \|\sigma - \tilde{\sigma}\|_{0,s',\Omega}$$
$$\leqslant \|\sigma - \tau_h\|_{0,s',\Omega} + \|\tau_h - \tilde{\sigma}\|_{0,s',\Omega} \quad \forall \tau_h \in X_h$$
$$\leqslant C \inf_{\tau_h \in X_h} \|\sigma - \tau_h\|_X.$$

Now, this last inequality, with (3.17), gives us

$$\|\sigma - \sigma_h\|_{0,s',\Omega}^{s'} + \int_\Omega (|\sigma| + |\sigma_h|)^{s'-2}|\sigma - \sigma_h|^2\,dx \leqslant C\,\|\sigma - \Pi_h\sigma\|_X^2.$$

Then, (3.13) is a consequence of this last estimation, (3.5), and the fact that (cf. Raviart & Thomas (1977), Nedelec (1980))

$$\|\nabla \cdot (\sigma - \Pi_h \sigma)\|_{0,\Omega} \leqslant C\, h\, |\nabla \cdot \sigma|_{1,\Omega}. \tag{3.19}$$

For the error estimation (3.14), we have (see the proof of Theorem 3.1)

$$\|P_h u - u_h\|_{0,\Omega} \leqslant C\, \|A(\sigma) - A(\sigma_h)\|_{0,s,\Omega},$$

and, using (3.16), (3.17) and (3.18), we obtain

$$\|P_h u - u_h\|_{0,\Omega} \leqslant C \inf_{\tau_h \in X_h} \|\sigma - \tau_h\|_X \leqslant C\, \|\sigma - \Pi_h \sigma\|_X.$$

Therefore, (3.14) is a consequence of this last inequality, (3.5), (3.19), the triangle inequality and the standard error estimate on the operator P_h. □

Let us close this paper by pointing out that the difficulties of extending the present analysis to the case of $s > 2$ come from the fact that (3.4) does not make sense since, in this case, $s' < 2$. One way to overcome these difficulties is to change the definition of the space X by using some additional regularity of the solution of problem (1.1). This problem will be the next step in our research.

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